Diode Parameters

1. **Dynamic Resistance:**

   It is the resistance the diode offers to the flow of AC when we connect it to a circuit with an AC voltage source as an active circuit element. Mathematically, the dynamic resistance is given as the ratio of the change in voltage applied across the diode to the resulting change in the current flowing through it.
   $R_{ac}=V_{ac}/I_{ac}$

2. **Static resistance**
   It is the resistance offered by the diode to the flow of DC through it when we apply a DC voltage to it. Mathematically the static resistance is expressed as the ratio of DC voltage applied across the diode terminals to the DC flowing through it.

   $R_{dc}=V_{dc}/I_{dc}$

3. **Knee Voltage or Cut-in voltage**

The voltage at which the forward diode current increases rapidly is known as Knee voltage or cut-in voltage. Knee voltage for germanium is 0.3V, and for silicon, it is 0.7V.

4. Peak Inverse Voltage

Peak inverse voltage is the maximum reverse bias voltage a p-n diode can withstand without breaking down.

5. Reverse saturation current:

A current flowing through a diode in the reverse-biased state is a reverse saturation current (IS).

The reverse saturation current in Si increases 100 % for each 100 C rise in temperature, i.e., approximately equal to 7 % for each 0 C rise.

6. Forward Voltage Drop: It is the voltage drop across the diode when it is forward conduction.

7. Maximum forward current:

The Maximum value of the forward current that a PN junction or diode can carry without damaging the device is called its Maximum Forward built-up Current. The manufacturer specifies the rating of a PN junction or a diode in its datasheet.

8. Reverse breakdown voltage:

When it is reverse biased and the voltage gradually increases to a certain value, there is an increase in reverse current. This is junction breakdown. The reverse voltage applied at this point is called the breakdown voltage of the PN junction diode.

9. Maximum power rating

Maximum power that a PN junction or diode can dissipate without damaging the device itself. Usually, it is specified by the manufacturer in, its datasheet. The power dissipated at the junction equals the product of the junction current and the voltage across the junction.